Q.16 Why is hyper parameter tuning important in deep learning? (CO2)

Q.17 What is the purpose of the flattening operations in CNNs? (CO2)

Q.18 What is the role of Tensor Flow in deep learning? (CO1)

Q.19 What is the difference between LSTMs and GRUs? (CO3)

Q.20 Differentiate between CNNs and RNNs? (CO1)

Q.21 How does padding affect the convolution operations? (CO2)

Q.22 What is text classification in NLP? (CO4)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Describe the key concept and significance of Deep Learning in real-wrold applications? (CO1)

Q.24 Discuss the role of activation functions in Convolutional Neural Networks (CNNs) and describe commonly used activation functions.(CO2)

Q.25 Explain the differences between image classification, object detection and image segmentation in computer vision. Provide examples of where each technique is used. (CO4)

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Roll No. ....

5th Sem/ Artificial Intelligence & Machine Learning
Subject : Deep Learning & its applications

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Which machine learning technique is commonly used in text classification? (CO4)

a) Convolutional Neural Networks (CNNs)

b) Support Vector Machines (SVMs)

c) Decision Trees

d) K-Nearest Neighbors (K-NN)

Q.2 The process commonly used to scale time series data before putting it into RNN? (CO3)

a) One-hot encoding

b) Image enhancement

c) Feature extraction

d) Normalization

Q.3 What is a common application of AI in finance? (CO4)

a) Automating customer support with chatbots

b) Analyzing social media sentiment

c) Fraud detection

d) Developing personalized meal plans

Q.4 Which activation function is most commonly used in CNNs? (CO2)

a) ReLU (Rectified Linear Unit)

b) Sigmoid

c) Tanh

d) Softmax

Q.5 Loss function commonly used for binary classification tasks is? (CO1)

a) Cross Entropy Loss

b) Mean Squared Error

c) Hinge Loss

d) Log-Loss

Q.6 Which of the following is a common challenge when training RNNs? (CO3)

a) Need for large convolutions filters

b) Over fitting due to large data sets

c) Vanishing gradient problem

d) Difficulty in processing images

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Which field users deep learning to discover potential new drugs? (CO4)

Q.8 (GANs) are composed of two neural networks: the generator and the_____. (CO1)

Q.9 Tensor Flow is primarily a data visulization library not a deep learning framework.(T/F) (CO2)

Q.10 Which RNN architecture mitigates the vanishing gradient problem? (CO3)

Q.11 Which technique is used for pixel-wise classification in images? (CO4)

Q.12 What activation function outputs values between 0 and 1? (CO1)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 What are some common applications of RNNs? (CO3)

Q.14 What is image segmentation in computer vision? (CO4)

Q.15 How is Deep Learning applied in healthcare? (CO1)